Where the Storks Go

Routes, funnels, sea crossings, winter quarters and fates of four migratory bird species in four open tracking data sets

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Abstract

I combine four open (CC0) animal tracking data sets, cleaned in the same way, to compare the migrations of 72 juvenile white storks *Ciconia ciconia* from eight populations (GPS, 2013 to 2014), 111 North American ospreys *Pandion haliaetus* (Argos, 1995 to 2002), 42 common cuckoos *Cuculus canorus* from a displacement experiment (Argos and GPS, 2015 to 2020) and 3 western marsh harriers *Circus aeruginosus* from Belgium (GPS, 2018 to 2022): 25,236 daily positions in all. Storks of the western populations crossed into Africa at the Strait of Gibraltar (13 of 18 that left home) and those of the eastern populations by the Turkish straits (21 of 23) and the Levant; none used the other's funnel. Measured as the share of distance between consecutive positions whose midpoint lies over open sea, storks flew 0.61% of their distance over water, against 3.7% for marsh harriers, 5.4% for cuckoos and 22.0% for ospreys. Autumn migrations took a median 61 days for storks (79 km a day), 51 for harriers and 23.5 for ospreys (170 km a day). The best single day reached 871 km for a marsh harrier. 30 of the 72 young storks (42%) were confirmed dead within the year, 16 before they had left home; among those that left, 5 of 7 Tunisian storks died, against 3 of 23 on the eastern route. A marsh harrier returned to within 2.6 km of the same winter site in Senegal in three winters.

Keywords: bird migration; animal tracking; white stork; osprey; common cuckoo; marsh harrier; flyways; Movebank; open data

1 Introduction

Long-distance migrants face the same problem with different bodies. White storks are heavy soaring birds that depend on thermals, which do not form over cold water; they are known to concentrate at the short sea crossings of the Strait of Gibraltar and the Bosporus and to split into western and eastern flyways around the Mediterranean (Flack et al., 2016). Ospreys are flapping fliers that fish and can cross wide stretches of sea (Martell et al., 2001); cuckoos are nocturnal migrants that cross the Mediterranean and the Sahara in long nonstop flights (Thorup et al., 2020b); marsh harriers soar and flap and fly over the desert by night.

Satellite and GPS tags have made these journeys visible bird by bird, and the Movebank Data Repository and Zenodo publish some of the resulting data openly. This paper does not add new tracking: it takes four such data sets, cleans them in one consistent way and compares routes, funnels, sea crossings, speeds, winter quarters and fates across species. It accompanies an interactive, hand-inked animated map (https://maps.tristanhotham.com/bird-migration/).

2 Data

White storks. The MPIO white stork lifetime tracking data (Flack et al., 2015): 72 juveniles fitted with e-obs GPS and accelerometer tags in the nest in 2013, in eight populations (south-west Germany, Spain, Tunisia, Poland, Greece, Russia, Armenia and Uzbekistan) and one bird tagged in South Africa in late 2012. The tags recorded a fix every five minutes in daylight; the GPS file has 1,584,192 rows. The reference data record which birds were confirmed dead. These data were collected for the comparison of migration costs across populations by Flack et al. (2016) and contribute to Kays et al. (2015).

Ospreys. Satellite (Argos) tracks of 111 adult and young ospreys tagged in the eastern, midwestern and western United States between 1995 and 2002 (Martell and Douglas, 2019; Martell et al., 2001), 86,383 rows. The data package notes that data for late 1998 and early 1999 were lost.

Cuckoos. Argos and Lotek GPS tracks of common cuckoos caught at Rybachy on the Curonian Spit (Thorup et al., 2020a,b). Some birds were released at Rybachy and others carried about 1,800 km east to Kazan before release; 23 control and 19 displaced birds have usable positions.

Marsh harriers. UvA-BiTS GPS tracks of western marsh harriers breeding near Antwerp, Belgium, 2018 to 2022 (INBO, 2023), 393,122 rows; the one common buzzard in the data set was left out. All four data sets are released under CC0. Coastlines, borders, lakes and towns are from Natural Earth (Natural Earth, 2026).

3 Methods

Cleaning. Rows not marked visible, without coordinates or flagged as outliers (by Movebank's Douglas Argos filter (Douglas et al., 2012) or by the data owners) were removed. Argos positions were kept only in location classes 3, 2 and 1, with stated errors under about 1.5 km: 19.1% of the osprey rows. A position was dropped if reaching it from the previous kept position needed more than 120 km/h (150 km/h for cuckoos) over a distance of more than 5 km, repeated until no such position remained. GPS tracks were thinned to one position per ten minutes (for the sea crossings) and per hour, and every track to one position a day, the one nearest 12:00 UTC (for Argos, the best location class first).

Distances and speeds. All distances are great-circle distances between positions and so lower bounds on the distance flown. A day's flight is the distance between consecutive daily positions 18 to 30 hours apart, scaled to 24 hours. The distance in a year is the sum of daily steps over the best-covered 365-day window of each bird followed for at least 330 days, with a position on at least 60% of the days.

Migrations. For each bird and year a summer anchor is the median daily position between 1 June and 31 July (or the tagging site for birds tagged in summer) and a winter anchor the median position between 1 December and 31 January (at least ten days with positions). An autumn migration runs from the last day within 100 km of the summer anchor to the first day within 200 km of the winter anchor, when the anchors are at least 500 km apart; a spring migration from the last day within 200 km of the winter anchor to the first day within 100 km of the next summer's anchor. Only migrations with a daily position on at least half the days and no gap over five days enter the speed comparisons.

Funnels. A stork crossed the Strait of Gibraltar if a position in Iberia was followed by one in Morocco; the Turkish straits if a position in Thrace and the Balkans was followed by one in Anatolia; the Levant if it had any position in Israel, Sinai or western Jordan (29 to 33.5°N, 34 to 36.6°E). A

Table 1: Young white storks by population: tagged, left home (at least 300 km), funnels crossed, with a winter position, median winter latitude (degrees) and confirmed dead.

Population	Tagged	Left	Gibraltar	Straits	Levant	Wintered	Lat.	Dead
SW Germany	13	9	6	0	0	7	34.3	6
Spain (Doñana)	11	9	7	0	0	2	15.4	4
Tunisia	9	7	0	0	0	4	12.2	7
Poland	4	4	0	3	3	2	-0.9	1
Greece (Evros Delta)	10	9	0	9	8	5	10.7	2
Russia (Moscow)	10	10	0	9	9	7	11.0	1
Armenia	8	4	0	0	0	5	33.1	5
Uzbekistan	6	0	0	0	0	1	40.4	4
South Africa	1	1	0	0	0	1	-33.8	0

stork that never went 300 km from its nest did not leave home.

Sea. Land is the union of Natural Earth's 1:10m countries. The share over the sea is the distance between consecutive positions (at most an hour apart for the GPS species, 24 hours for ospreys and 72 for cuckoos) whose midpoint lies over the sea, as a share of all such distance. A crossing is a run of positions or sampled points over the sea between two on land; for the satellite tags it runs along the straight line between positions.

Winter quarters. For each species the December and January daily positions (at most 40 per bird) were smoothed with a Gaussian kernel and the smallest areas holding 50% and 90% of the density drawn.

4 Results

4.1 Flyways and funnels

Table 1 and Figure 2 show the stork populations. Of the 18 storks from Germany and Spain that left home, 13 crossed at Gibraltar; the others wintered in Spain or France or died first. Of the 23 from Poland, Greece and Moscow that left, 21 crossed the Turkish straits and 20 passed through the Levant. No stork used the other side's funnel. 7 Tunisian birds crossed the Sahara directly; the 4 Armenian birds that left went south through Turkey, Iraq and Iran, and none of the Uzbek storks went more than 141 km from home. Figure 1 shows the four species side by side.

4.2 Around the sea or across it

Storks flew 0.61% of their distance over the sea (2,360 of 389,086 km), and their longest run over water was 82 km (Figure 3). Ospreys flew 22.0% over the sea, about 36 times the storks' share; 4 made crossings of 300 km or more, mostly over the Caribbean between the Greater Antilles and northern South America (median 728 km). One osprey, KD, left Hispaniola, spent 82 hours over the Caribbean and returned to Haiti after 859 km. 4 cuckoos have straight lines between positions from southern Europe across the Mediterranean, the longest about 1,027 km; one young cuckoo went from Poland to Chad, 4,347 km, in 4.8 days. The harriers' longest crossing was 123 km.

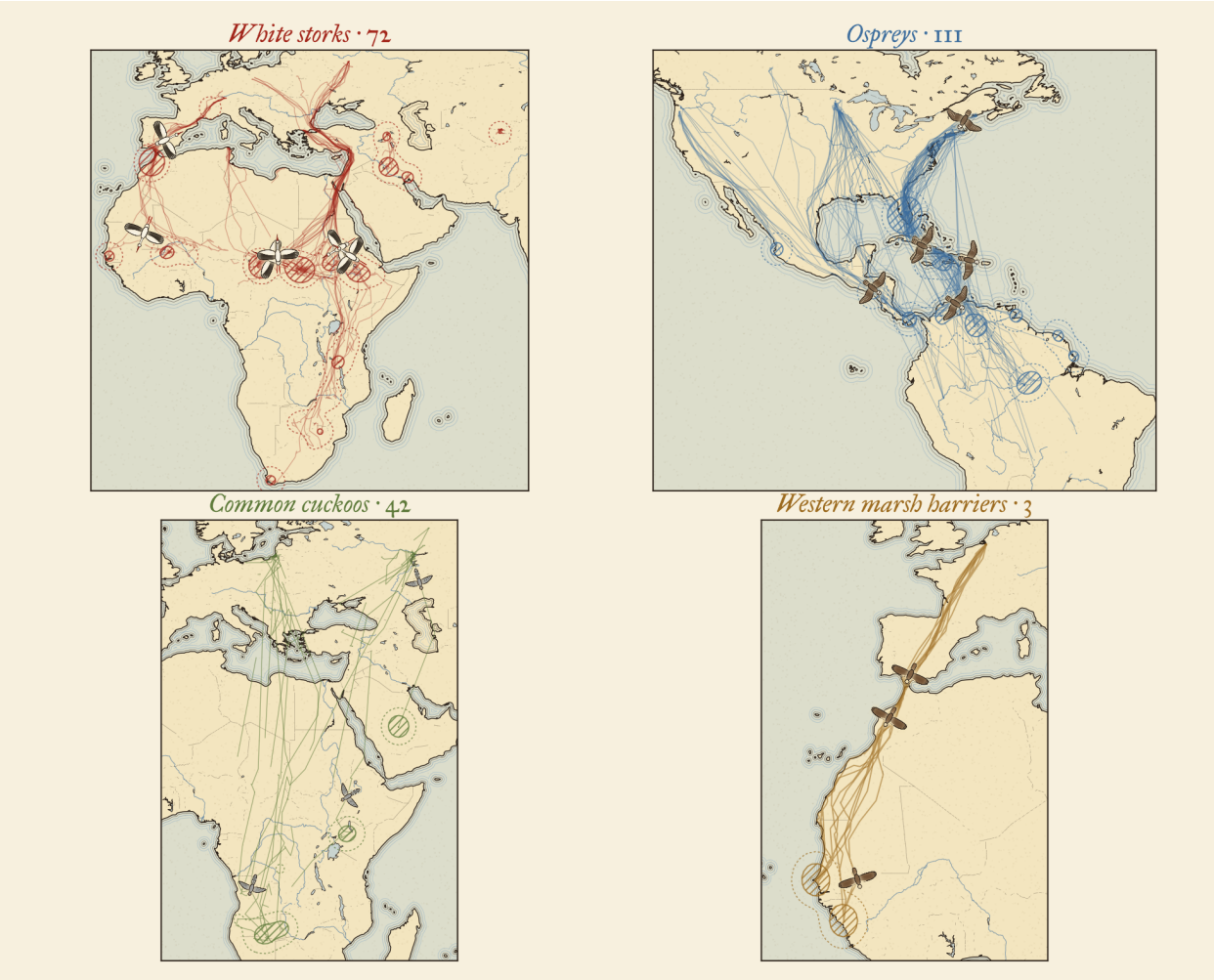


Figure 1: Daily tracks of the four species (faint lines), with pen-drawn birds on sample autumn migrations and the winter quarters hatched (50% of the December and January density; dashed: 90%).

Table 2: Well-covered migrations: number, and medians of the distance between daily positions (km), the straight distance between anchors (km), days, km a day and days with more than 50 km of flight.

Species	Season	n	Path	Straight	Days	km/day	Travel days
White stork	autumn	17	4,829	3,298	61	79	16
White stork	spring	12	3,978	3,114	52	76	18
Osprey	autumn	22	5,301	3,975	24	170	6
Osprey	spring	8	4,977	3,786	23	237	3
Western marsh harrier	autumn	5	4,639	4,492	51	89	19
Western marsh harrier	spring	3	4,719	4,491	36	135	22



Figure 2: Young white storks by population, with the three funnels and the last positions of birds found dead (crosses).

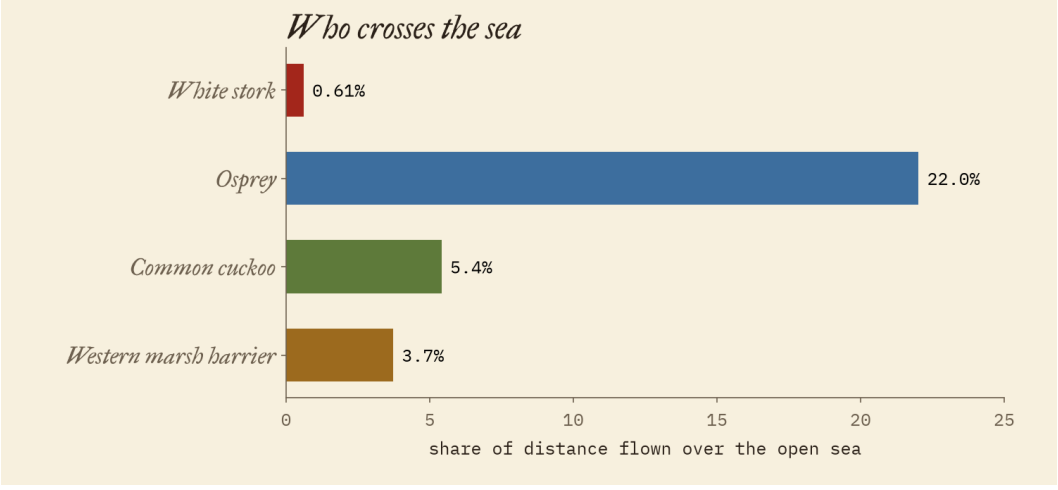


Figure 3: Share of distance flown over the open sea, by species.

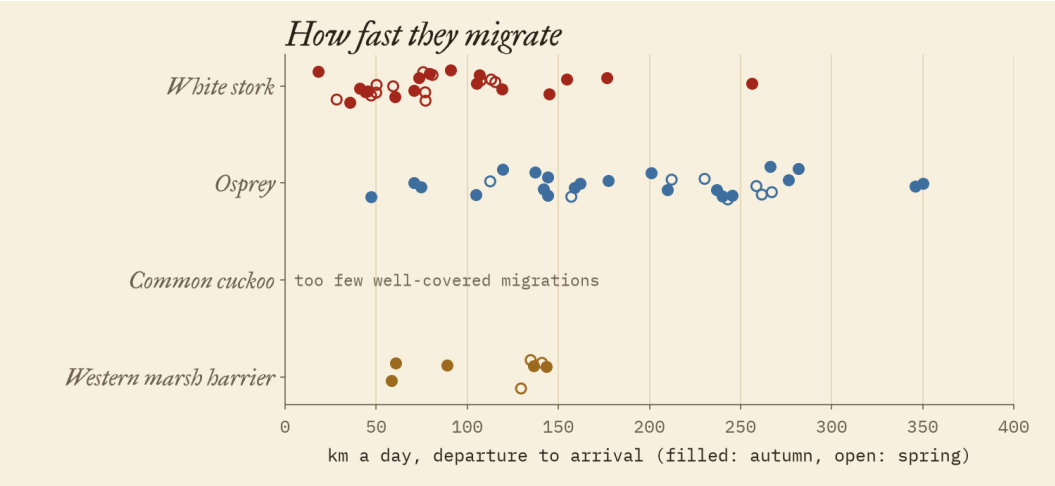


Figure 4: Speed of each well-covered migration (distance between daily positions over days from departure to arrival).

4.3 Speed and distance

Table 2 and Figure 4 summarise the well-covered migrations. Stork autumn migrations covered a median 4,829 km in 61 days, 79 km a day including stops, with a detour of 18% over the straight line between anchors; ospreys went 2.1 times as fast. Cuckoo tracks were too sparse for this measure. The best day of the median migrating stork was 371 km and the best of all 558 km; ospreys reached 730 km and a marsh harrier 871 km, from central Spain to Morocco with flight continuing after dark. Over a year, 17 storks covered a median 12,724 km between daily positions and at most 23,680 km; between hourly positions the stork tracks are 41% longer.

4.4 Winter quarters

18 of the 34 storks with a winter position spent December and January in the Sahel belt (5 to 20°N), 5 in East and southern Africa and 5 in Europe and Morocco (Figure 5). The farthest, DER AR445 from Moscow, reached Zimbabwe, 9,186 km from its nest. Of 65 ospreys tracked south of 25°N, 38 reached South America; 13 went no farther than Mexico, 9 of them from the western population. The two long-tracked harriers wintered at the same sites every year: Lilla's winter centres near Dakar in three winters lay within 2.6 km of each other, and Lillo's in Guinea 0.8 km apart (Figure 6).

4.5 Fates

30 of the 72 storks were confirmed dead (42%); 16 of them never went 300 km from home and 15 died before 1 October 2013 (Figure 7). Among birds that left home, deaths were 3 of 23 on the eastern route, 4 of 18 on the western route and 5 of 7 for Tunisian birds crossing the Sahara. These small numbers are descriptive only; Flack et al. (2016) analyse the costs of the routes in detail. The osprey records note 8 deaths, 3 of them probably on floating debris at sea.

4.6 The displaced cuckoos

Of the displaced cuckoos whose first 500 km from Kazan could be followed, 8 of 9 headed south-west (median bearing 201 degrees), against a median 176 degrees for 14 control birds (Figure 8).

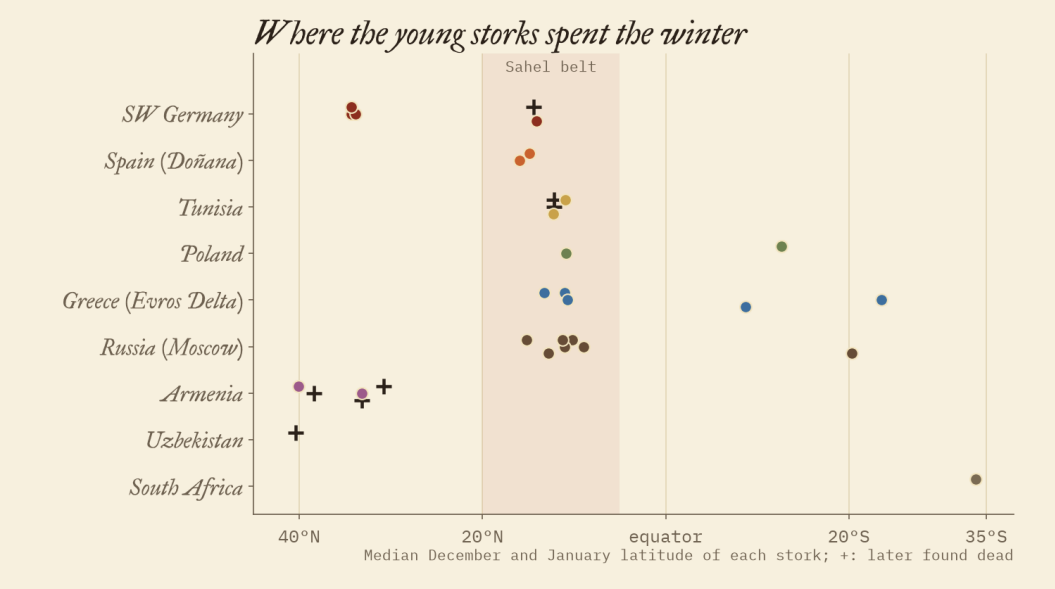


Figure 5: Median December and January latitude of each stork, by population; crosses mark birds later found dead.

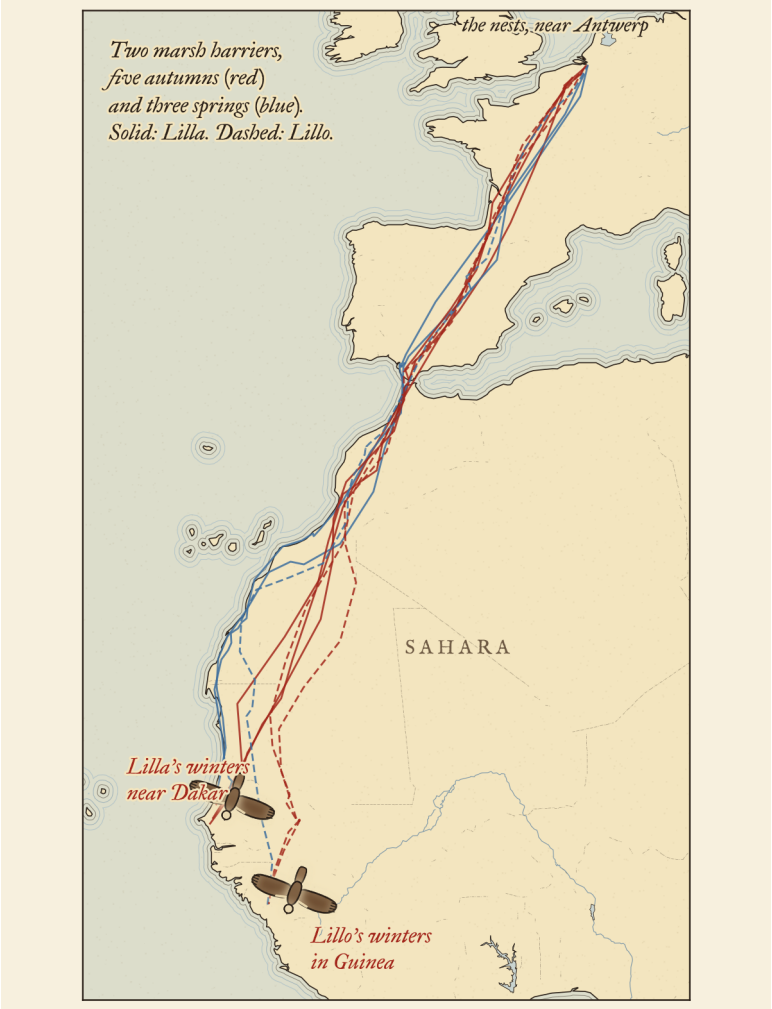


Figure 6: The migrations of two marsh harriers from Antwerp: autumn in red, spring in blue.

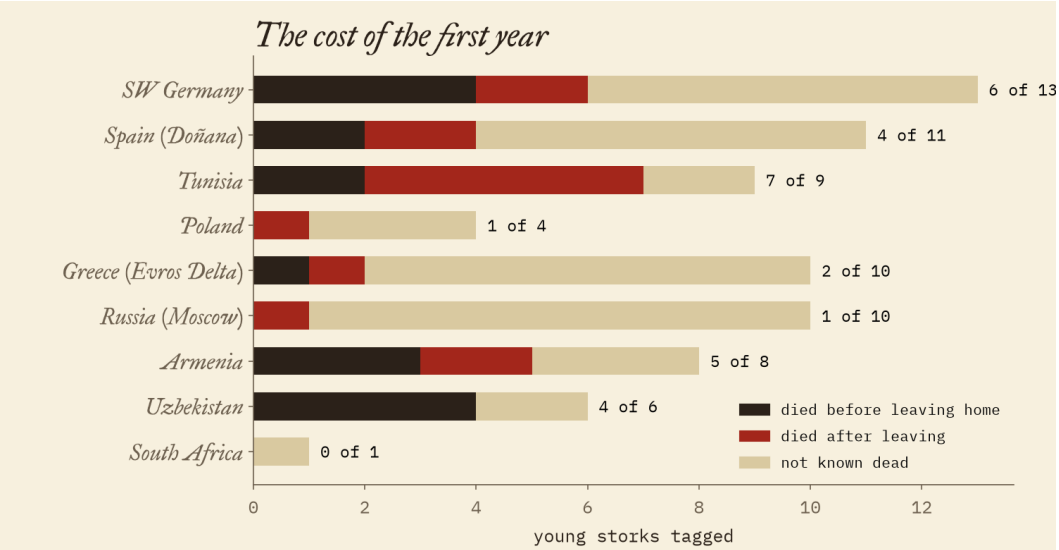


Figure 7: Young storks by population: died before leaving home, died after leaving, and not known to have died.

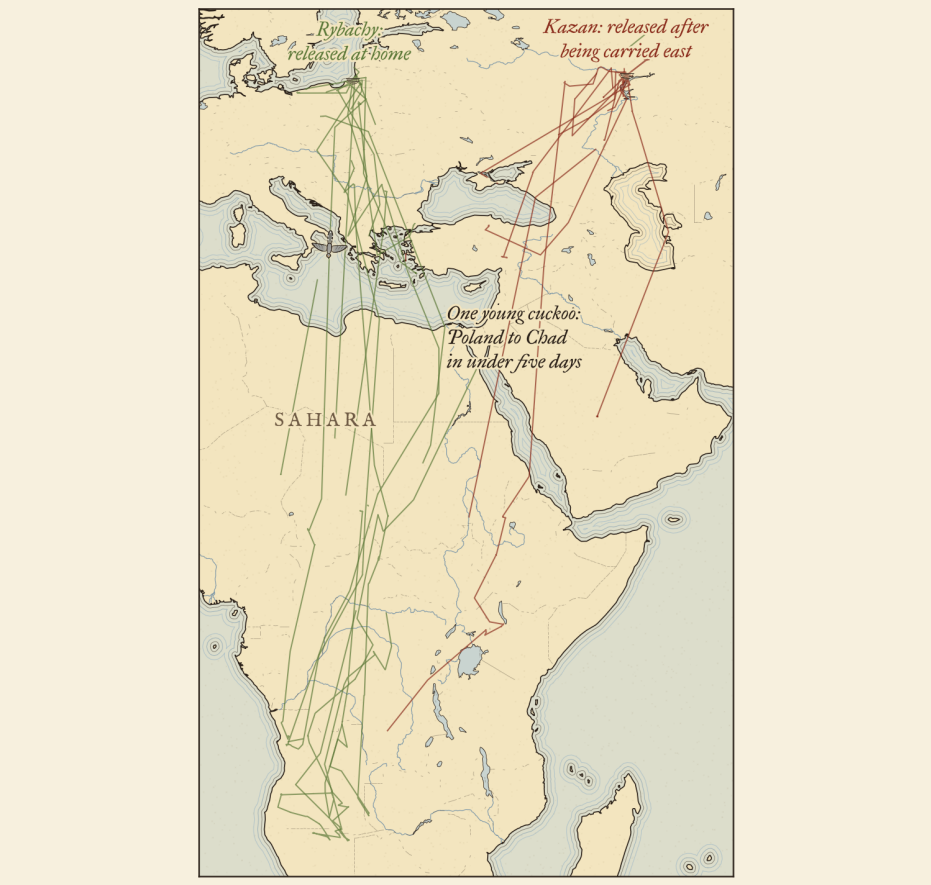


Figure 8: Cuckoos released at Rybachy (green) and after displacement to Kazan (red).

Tags were short-lived (median 53 and 26 days), so the end points are not known for most; Thorup et al. (2020b) give the full analysis.

5 Limitations

First, the samples are small and specific: one cohort of young storks in one year, ospreys of the late 1990s and three harriers; they describe these birds, not their populations. Second, satellite tags give a few positions a day with gaps of days, so osprey and cuckoo tracks are straight lines between sparse points; sea crossings and speeds are approximate for them and distances are lower bounds for all species. Third, the anchors and thresholds that define migrations, funnels and leaving home are choices; others would shift individual birds. Fourth, a tag that stops is not a dead bird: only deaths confirmed by the data owners are counted. Fifth, coastlines at 1:10m mislabel some positions on the shore as at sea. Finally, the cuckoo data come from an experiment, and the displaced birds' routes are not natural ones.

6 Conclusion

Put on one map, the four data sets show four ways of crossing the same barriers. The soaring storks treat the Mediterranean as a wall and squeeze through its narrowest gaps, and the young storks of Europe divide cleanly between Gibraltar and the Bosporus. Ospreys and cuckoos cross open sea, ospreys by day over the Caribbean and cuckoos by night over the Mediterranean and the Sahara. Harriers return year after year to the same few kilometres of West Africa. And the first year is costly: four young storks in ten did not survive it.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/bird-migration/. Data and code are at https://github.com/Tristan-Hotham/bird-migration and archived with this paper at https://doi.org/10.5281/zenodo.24043676. Derived data are released under CC BY 4.0 and code under the MIT licence; the tracking data are CC0.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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A The young storks

Table 3: The 72 young storks: first and last daily position, distance between daily positions (km), farthest distance from the nest (km), winter country and confirmed deaths.

Stork	Population	First	Last	km	Farthest	Winter	Fate
BraveHeart (DER AP926)	Armenia	2013-06-16	2014-01-22	2,290	1,134	Iran	dead
DER AP921	Armenia	2013-06-16	2013-09-26	142	29		dead
DER AP922	Armenia	2013-06-16	2013-12-13	960	236	Turkey	dead
DER AP923	Armenia	2013-06-16	2013-11-25	952	535		
DER AP924	Armenia	2013-06-16	2014-03-04	1,015	790	Iraq	dead
DER AP925	Armenia	2013-06-17	2014-02-12	87	19	Armenia	
DER AP927	Armenia	2013-06-16	2013-09-11	94	27		dead
DER AP928	Armenia	2013-06-17	2014-06-15	3,854	866	Iraq	
Dimitris (GRA51162)	Greece (Evros Delta)	2013-07-03	2013-08-24	128	111		dead
Hercules (GRA51167)	Greece (Evros Delta)	2013-07-02	2013-09-02	638	513		
Lina (GRA51160)	Greece (Evros Delta)	2013-07-02	2013-09-14	3,672	2,852		dead
Nenni (GRA51165)	Greece (Evros Delta)	2013-07-02	2014-04-01	10,203	3,353	Sudan	
Pelopidas (GRA51164)	Greece (Evros Delta)	2013-07-02	2014-07-02	18,982	7,212	South Africa	
Petros (GRA51163)	Greece (Evros Delta)	2013-07-02	2014-04-15	8,605	5,623	Tanzania	
Silvia (GRA51159)	Greece (Evros Delta)	2013-07-02	2014-07-02	14,164	3,719	Sudan	
Sophokles (GRA51157)	Greece (Evros Delta)	2013-07-02	2013-10-28	4,448	3,290		
Spithas (GRA51158)	Greece (Evros Delta)	2013-07-02	2014-02-09	7,276	3,570	Ethiopia	
Vasilia (GRA51161)	Greece (Evros Delta)	2013-07-02	2013-08-22	3,509	3,048		
PLG 5P305	Poland	2013-07-16	2013-11-28	12,283	8,884		
PLG 5P307	Poland	2013-07-16	2013-11-07	1,646	1,501		dead
PLG 5P311	Poland	2013-07-16	2014-07-16	18,840	7,763	Zambia	
PLG 5P312	Poland	2013-07-16	2014-07-16	15,916	4,754	Sudan	
DER AR436	Russia (Moscow)	2013-07-12	2013-11-06	7,052	5,135		
DER AR437	Russia (Moscow)	2013-07-12	2013-09-09	633	638		dead
DER AR438	Russia (Moscow)	2013-07-12	2014-06-19	9,778	5,381	Ethiopia	
DER AR439	Russia (Moscow)	2013-07-12	2013-12-03	9,364	6,750		
DER AR440	Russia (Moscow)	2013-07-12	2014-07-12	19,089	5,187	Sudan	
DER AR441	Russia (Moscow)	2013-07-12	2014-01-30	12,247	6,896	Sudan	
DER AR442	Russia (Moscow)	2013-07-12	2014-03-29	7,258	5,263	Chad	
DER AR444	Russia (Moscow)	2013-07-12	2014-07-12	22,004	8,015	Ethiopia	
DER AR445	Russia (Moscow)	2013-07-12	2014-07-12	23,680	9,186	Zimbabwe	
DER AR446	Russia (Moscow)	2013-07-12	2014-07-12	12,724	4,610	Sudan	
Benjamin (DER AN867)	SW Germany	2013-07-15	2013-08-09	0	0		dead
Cherry (DER AN859)	SW Germany	2013-06-24	2013-09-02	1,348	1,166		dead
Coco (DER AN862)	SW Germany	2013-06-24	2014-01-25	5,540	3,918	Mali	dead
Eamy (DER AL585)	SW Germany	2013-06-24	2014-06-24	5,016	1,979	Morocco	
Kakau (DER AN864)	SW Germany	2013-06-24	2013-12-21	568	396	France	
Karamell (DER AL582)	SW Germany	2013-06-24	2014-06-24	5,767	1,983	Morocco	
Keks (DER AN860)	SW Germany	2013-06-25	2013-08-22	1,927	1,770		
Pieps (DER AL586)	SW Germany	2013-06-25	2013-08-23	146	97		dead
Sierit (DER AN858)	SW Germany	2013-06-24	2014-06-24	4,038	2,028	Morocco	
Swag (DER AL583)	SW Germany	2013-06-24	2013-07-02	0	0		dead
Willi (DER AN863)	SW Germany	2013-06-24	2013-08-08	112	23		dead
Yolo (DER AN861)	SW Germany	2013-06-24	2014-06-24	12,050	4,451	Senegal	
Zozu (DER AL581)	SW Germany	2013-06-24	2014-06-24	4,138	1,979	Morocco	
ZAC 9A69901	South Africa	2012-11-30	2013-04-07	2,812	2,246	South Africa	
C64W	Spain (Doñana)	2013-06-11	2013-07-20	3	21		dead
C666	Spain (Doñana)	2013-07-15	2013-11-04	8,312	2,916		dead
C703	Spain (Doñana)	2013-06-11	2013-07-09	3	22		dead
C708	Spain (Doñana)	2013-06-11	2014-04-25	4,512	3,536		dead
C70V	Spain (Doñana)	2013-06-11	2013-11-17	4,327	2,735		
C711	Spain (Doñana)	2013-06-11	2014-03-07	5,399	2,594	Mali	
C7VH	Spain (Doñana)	2013-06-12	2013-11-03	3,153	2,514		

Stork	Population	First	Last	km	Farthest	Winter	Fate
C7W0	Spain (Doñana)	2013-06-12	2013-08-01	397	313		
C7X7	Spain (Doñana)	2013-06-12	2014-06-11	8,906	2,714	Mali	
C8WN	Spain (Doñana)	2013-06-12	2013-07-24	488	465		
CAMF	Spain (Doñana)	2013-06-17	2013-07-25	746	741		
DER AP801	Tunisia	2013-06-08	2013-07-30	271	202		dead
DER AP802	Tunisia	2013-06-08	2013-11-03	2,792	2,574		dead
DER AP803	Tunisia	2013-06-08	2014-04-04	7,849	3,193	Chad	dead
DER AP804	Tunisia	2013-06-08	2014-02-16	5,521	2,862	Chad	dead
DER AP805	Tunisia	2013-06-08	2013-12-04	4,026	3,165		dead
DER AP806	Tunisia	2013-06-08	2013-08-20	335	228		dead
DER AP807	Tunisia	2013-06-08	2013-11-02	1,340	1,221		dead
DER AP808	Tunisia	2013-06-08	2014-06-08	11,058	3,501	Sudan	
DER AP809	Tunisia	2013-06-08	2014-06-08	13,016	3,361	Sudan	
Dilya	Uzbekistan	2013-06-01	2014-06-01	1,059	90		dead
Grigori	Uzbekistan	2013-06-01	2014-02-10	729	64		dead
Halim	Uzbekistan	2013-06-01	2013-12-24	993	95	Uzbekistan	dead
Mansur	Uzbekistan	2013-06-01	2013-08-09	180	26		dead
Sayyid	Uzbekistan	2013-06-01	2014-06-01	1,068	63		
Shahlo	Uzbekistan	2013-06-01	2014-06-01	1,280	141		